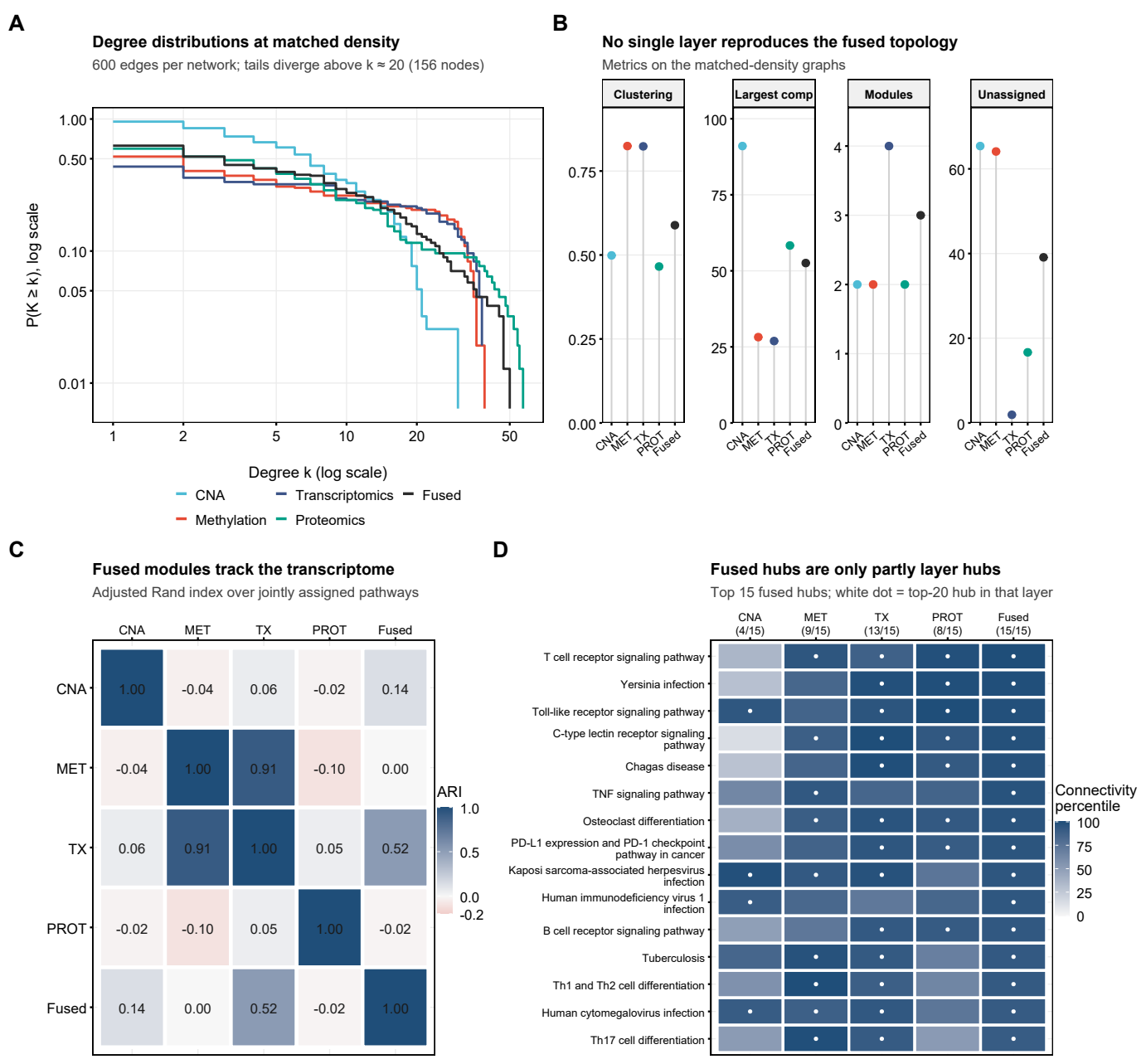

A
Degree distributions at matched density
600 edges per network; tails diverge above k ≈ 20 (156 nodes)
P(K ≥ k), log scale
1.00
0.50
0.10
0.05
0.01
1
2
5
10
20
50
Degree k (log scale)
CNA
Transcriptomics
Fused
Methylation
Proteomics
B
No single layer reproduces the fused topology
Metrics on the matched-density graphs
Clustering
Largest comp
Modules
Unassigned
CNA
MET
TX
PROT
Fused
C
Fused modules track the transcriptome
Adjusted Rand index over jointly assigned pathways
CNA
MET
TX
PROT
Fused
1.00
-0.04
0.06
-0.02
0.14
-0.04
1.00
0.91
-0.10
0.00
0.06
0.91
1.00
0.05
0.52
-0.02
-0.10
0.05
1.00
-0.02
0.14
0.00
0.52
-0.02
1.00
ARI
1.0
0.5
0.0
-0.2
D
Fused hubs are only partly layer hubs
Top 15 fused hubs; white dot = top-20 hub in that layer
CNA (4/15)
MET (9/15)
TX (13/15)
PROT (8/15)
Fused (15/15)
T cell receptor signaling pathway
Yersinia infection
Toll-like receptor signaling pathway
C-type lectin receptor signaling pathway
Chagas disease
TNF signaling pathway
Osteoclast differentiation
PD-L1 expression and PD-1 checkpoint pathway in cancer
Kaposi sarcoma-associated herpesvirus infection
Human immunodeficiency virus 1 infection
B cell receptor signaling pathway
Tuberculosis
Th1 and Th2 cell differentiation
Human cytomegalovirus infection
Th17 cell differentiation
Connectivity percentile
100
75
50
25
0